

NORMAL DISTRIBUTION

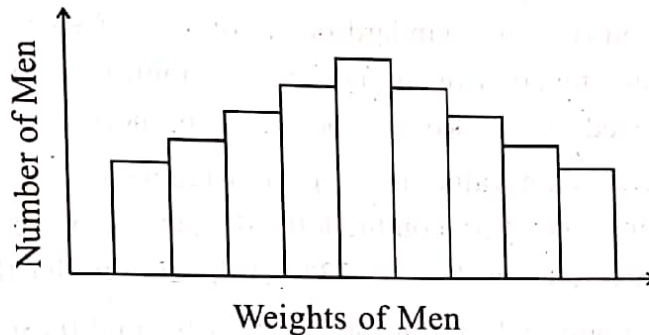
The normal law of error stands out in the mankind as one of the broadest generalisations of natural philosophy. It serves as the guiding instrument in researches in the physical and social sciences and in engineering, medicine and agriculture. It is an indispensable tool for the analysis and interpretation of the basic data obtained by Observation and experience.

-W.J. Youden

Normal distribution is applicable in the following situations:

1. Life of items subjected to wear and tear like tyres, batteries, bulbs, currency notes, etc.
2. Length and diameter of certain products like pipes, screws and discs.
3. Height and weight of baby at birth.
4. Aggregate marks obtained by students in an examination.
5. Weekly sales of an item in a store.

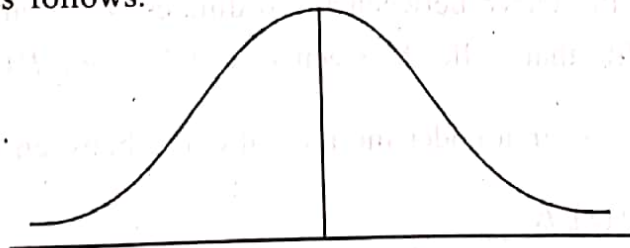
The most common pattern of distribution of a continuous variable, found in nature, is of the following type:



Distribution of Weights of Men

Maximum number of men will be around the average weight, and as one goes to the left or right, the number of persons will keep on reducing. In fact, even other human characteristics like age, intelligence, etc. follow this pattern or distribution.

This type of distribution was discovered by Karl Gauss. For this reason, it is called Gaussian Distribution. But, it is found so often in real life that it is called Normal Distribution. The name which is commonly used. If the curve is drawn in a continuous manner, it would look as follows.



Shape of Normal Distribution

3.10 NORMAL DISTRIBUTION

The distributions discussed so far are discrete distributions *i.e.*, distributions in which the variate can take only integral values. Examples of such distributions are the numbers of defective items manufactured in a factory, the numbers of persons suffering from a disease, etc. Now we shall consider a continuous distribution, namely the *Normal*

Distribution. A continuous distribution is a distribution in which the variate can take all values within a given range. Examples of continuous distributions are the heights of persons, the speed of a vehicle, etc.

The Normal distribution was first discovered by English Mathematician De-Moivre (1667-1745) in 1733 and further refined by French Mathematician Laplace (1749-1827) in 1774 and independently by Karl Friedrich Gauss (1777-1855). Normal distribution is also known as Gaussian Distribution. It is another limiting form of the Binomial Distribution for large values of n when neither p nor q is very small. The Normal Distribution is, therefore, derived from the Binomial Distribution by increasing the number of trials indefinitely.

Definition : A random variable X is said to have a Normal Distribution, if its density function or probability distribution is given by [JNTU (H) Dec. 2009 (Set No. 2)]

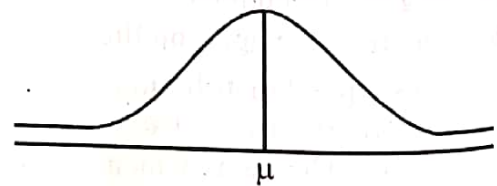
$$f(x; \mu, \sigma) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty, -\infty < \mu < \infty; \sigma > 0 \quad \dots (1)$$

where μ is the Mean and σ is the Standard deviation of x . As can be seen, the function, called probability density function of the normal distribution, depends on two values μ and σ . These are referred as the two parameters of the normal distribution.

The curve has maximum value at $x = \mu$ and tapers off on either side but never touches the horizontal line. The curve on the left side goes upto $-\infty$, and on the right side it goes upto $+\infty$. However, as much as 99.73% of the area under the curve lies between $(\mu - 3\sigma)$ and $(\mu + 3\sigma)$ and only 0.277% of the area lies beyond these points.

The random variable X is then said to be a normal random variable or normal variate. The curve representing the Normal Distribution (1) is called the normal curve and the total area bounded by the curve and the x -axis is one.

$$\text{i.e., } \int_{-\infty}^{\infty} f(x) dx = 1$$



The area under the curve between the ordinates $x = a$ and $x = b$, where $a < b$, represents the probability that x lies between a and b . i.e., $P(a < x < b)$.

Thus $P(a < x < b) =$ area under the normal curve between the vertical lines $x = a$ and $x = b$, which is $\int_a^b f(x) dx$.

Note : A random variable X with mean μ and variance σ^2 and following the probability law (1) is expressed by $X \sim N(\mu, \sigma^2)$.

3.11 NORMAL DISTRIBUTION AS A LIMITING FORM OF BINOMIAL DISTRIBUTION

[JNTU (A) Dec. 2009, (K) Nov. 2011 (Set No. 1)]

Normal distribution is a limiting case of the Binomial distribution under the following conditions :

- (i) n , the number of trials is indefinitely large i.e., $n \rightarrow \infty$.
 (ii) neither p nor q is very small.

The probability of x successes in a series of n independent trials is

$$p(x) = {}^nC_x p^x q^{n-x}; \quad x = 0, 1, 2, \dots, n \quad \dots (1)$$

Let us now consider the variate

$$Z = \frac{X - np}{\sqrt{npq}}, \quad X = 0, 1, 2, \dots, n \quad \dots (2)$$

When $X = 0, Z = \frac{-np}{\sqrt{npq}} = -\sqrt{\frac{np}{q}}$ and

When $X = n, Z = \frac{n - np}{\sqrt{npq}} = \frac{n(1-p)}{\sqrt{npq}}$

$$= \frac{nq}{\sqrt{npq}} = \sqrt{\frac{nq}{p}}$$

Thus in the limit as $n \rightarrow \infty$, Z takes the values from $-\infty$ to ∞ . Hence the distribution of X will be a continuous distribution over the range $-\infty$ to ∞ .

We want the limiting form of (1) under the above conditions.

Hence $\lim_{n \rightarrow \infty} p(x) = \lim_{n \rightarrow \infty} \frac{n!}{x!(n-x)!} p^x q^{n-x}$

Using Stirling's approximation for $n!$ as $n \rightarrow \infty$, $\lim_{n \rightarrow \infty} n! \approx \sqrt{2\pi} e^{-n} n^{n+\frac{1}{2}}$, we get

$$\begin{aligned} \lim_{n \rightarrow \infty} p(x) &= \lim_{n \rightarrow \infty} \left[\frac{\sqrt{2\pi} e^{-n} n^{n+\frac{1}{2}} \cdot p^x q^{n-x}}{\left(\sqrt{2\pi} e^{-x} x^{x+\frac{1}{2}} \right) \left(\sqrt{2\pi} e^{-(n-x)} (n-x)^{n-x+\frac{1}{2}} \right)} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2\pi}} \cdot \frac{n^{n+\frac{1}{2}}}{x^{x+\frac{1}{2}} (n-x)^{n-x+\frac{1}{2}}} \cdot p^x q^{n-x} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} \cdot \frac{(np)^{x+\frac{1}{2}} (nq)^{n-x+\frac{1}{2}}}{x^{x+\frac{1}{2}} (n-x)^{n-x+\frac{1}{2}}} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} \cdot \left(\frac{np}{x} \right)^{x+\frac{1}{2}} \cdot \left(\frac{nq}{n-x} \right)^{n-x+\frac{1}{2}} \right] \quad \dots (3) \end{aligned}$$

Now, substitute $N = \left(\frac{x}{np} \right)^{x+\frac{1}{2}} \cdot \left(\frac{n-x}{nq} \right)^{n-x+\frac{1}{2}}$

Then (3) becomes $\lim_{n \rightarrow \infty} p(x) = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} \cdot \frac{1}{N} \quad \dots (4)$

From (2), we have

$$X = np + Z \cdot \sqrt{npq} \quad \dots (4.1)$$

$$\Rightarrow \frac{X}{np} = 1 + Z \cdot \sqrt{\frac{q}{np}} \quad \dots (4.2)$$

$$\begin{aligned} \text{Also } n - X &= n - np - Z \cdot \sqrt{npq} = n(1 - p) - Z \cdot \sqrt{npq} \\ &= nq - Z \cdot \sqrt{npq} \end{aligned} \quad \dots (4.3)$$

$$\therefore \frac{n - X}{nq} = 1 - Z \cdot \sqrt{\frac{p}{nq}} \quad \dots (4.4)$$

$$\begin{aligned} \text{Now } \log N &= \left(x + \frac{1}{2}\right) \log\left(\frac{x}{np}\right) + \left(n - x + \frac{1}{2}\right) \cdot \log\left(\frac{n - x}{nq}\right) \\ &= \left(np + Z \cdot \sqrt{npq} + \frac{1}{2}\right) \cdot \log\left(1 + Z \cdot \sqrt{\frac{q}{np}}\right) + \left(nq - Z \cdot \sqrt{npq} + \frac{1}{2}\right) \cdot \log\left(1 - Z \cdot \sqrt{\frac{p}{nq}}\right) \\ &\quad \text{using (4.1), (4.2), (4.3), (4.4)} \\ &= \left(np + Z \cdot \sqrt{npq} + \frac{1}{2}\right) \left[Z \cdot \sqrt{\frac{q}{np}} - \frac{1}{2} Z^2 \left(\frac{q}{np}\right) + \frac{1}{3} Z^3 \left(\frac{q}{np}\right)^{3/2} - \dots \right] \\ &\quad + \left(nq - Z \cdot \sqrt{npq} + \frac{1}{2}\right) \left[-Z \sqrt{\frac{p}{nq}} - \frac{1}{2} Z^2 \left(\frac{p}{nq}\right) - \frac{1}{3} Z^3 \left(\frac{p}{nq}\right)^{3/2} - \dots \right] \\ &= \left[\left\{ Z \sqrt{npq} - \frac{1}{2} q Z^2 + \frac{1}{3} Z^3 \frac{q^{3/2}}{\sqrt{np}} + Z^2 q - \frac{1}{2} Z^3 \frac{q^{3/2}}{\sqrt{np}} + \frac{1}{2} Z \cdot \sqrt{\frac{q}{np}} - \frac{1}{4} Z^2 \cdot \frac{q}{np} + \dots \right\} \right. \\ &\quad \left. + \left\{ -Z \sqrt{npq} - \frac{1}{2} Z^2 p - \frac{1}{3} Z^3 \frac{p^{3/2}}{\sqrt{nq}} + Z^2 p + \frac{1}{2} Z^3 \frac{p^{3/2}}{\sqrt{nq}} - \frac{1}{2} Z \cdot \sqrt{\frac{p}{nq}} - \frac{1}{4} Z^2 \frac{p}{nq} + \dots \right\} \right] \\ \text{i.e., } \log N &= \left[-\frac{1}{2} Z^2 (p + q) + Z^2 (p + q) + \frac{Z}{2\sqrt{n}} \left\{ \sqrt{\frac{q}{p}} + \sqrt{\frac{p}{q}} \right\} + O\{n^{-1/2}\} \right] \\ &= \frac{Z^2}{2} + O(n^{-1/2}) \rightarrow \frac{Z^2}{2} \text{ as } n \rightarrow \infty \end{aligned}$$

$$\therefore \lim_{n \rightarrow \infty} \log N = \frac{Z^2}{2} \Rightarrow \lim_{n \rightarrow \infty} N = e^{Z^2/2}$$

Now, from (4) we have

$$\lim_{n \rightarrow \infty} p(x) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} e^{-Z^2/2}$$

$$\text{or } P(z \leq Z \leq z + dz) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{npq}} e^{-Z^2/2} \quad \dots (5)$$

$$\text{But } Z = \frac{X - np}{\sqrt{npq}} = \frac{x - \mu}{\sigma}$$

To obtain the distribution of X , we put $Z = \frac{x - \mu}{\sigma}$ in (5). We get

$$P(x \leq X \leq x + dx) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

3.12 CONSTANTS OF NORMAL DISTRIBUTION

1. Mean of Normal Distribution

[JNTU 2006S, (H) Dec. 2009 (Set No.2)]

Consider the Normal Distribution with b, σ as the parameters.

$$\text{Then } f(x; b, \sigma) = \frac{e^{-\frac{(x-b)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$$

The mean $\mu = E(X)$ is given by

$$\begin{aligned} \mu &= \int_{-\infty}^{\infty} x f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-\frac{1}{2}\left(\frac{x-b}{\sigma}\right)^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma z + b) e^{-\frac{z^2}{2}} dz \quad \left[\text{Putting } z = \frac{x-b}{\sigma} \text{ so that } dz = \frac{dx}{\sigma} \right] \\ &= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz + \frac{b}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \\ &= 0 + \frac{b}{\sqrt{2\pi}} \cdot 2 \int_0^{\infty} e^{-\frac{z^2}{2}} dz \quad \left[\because z e^{-\frac{z^2}{2}} \text{ is odd function and } e^{-\frac{z^2}{2}} \text{ is even function} \right] \\ &= \frac{2b}{\sqrt{2\pi}} \cdot \frac{\sqrt{\pi}}{\sqrt{2}} \quad \left[\because \int_0^{\infty} e^{-\frac{x^2}{2}} dx = \sqrt{\frac{\pi}{2}} \right] \\ &= b \end{aligned}$$

$\therefore$ Mean, $\mu = b$

2. Variance of Normal Distribution

By definition,

$$\begin{aligned} \text{Variance} &= E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - b)^2 e^{-\frac{1}{2}\left(\frac{x-b}{\sigma}\right)^2} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\sigma z)^2 e^{-\frac{z^2}{2}} dz \quad \left[\text{Putting } z = \frac{x-b}{\sigma} \text{ so that } dz = \frac{dx}{\sigma} \right] \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-\frac{z^2}{2}} dz \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \cdot 2 \int_0^{\infty} z^2 e^{-\frac{z^2}{2}} dz \quad [\because \text{Integrand is even function}] \\
&= \frac{2\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} 2t e^{-t} \cdot \frac{dt}{\sqrt{2t}} \quad \left[\text{Putting } \frac{z^2}{2} = t \text{ so that } dz = \frac{dt}{\sqrt{2t}} \right] \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} e^{-t} \sqrt{t} dt = \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} e^{-t} t^{\frac{3}{2}-1} dt = \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \sigma^2
\end{aligned}$$

Hence variance = σ^2

Thus the Standard deviation of the Normal Distribution is σ .

3. Mode of Normal Distribution

[JNTU 2004 S]

Mode is the value of x for which $f(x)$ is maximum. That is, mode is the solution of

$$f'(x) = 0 \text{ and } f''(x) < 0$$

By definition, we have

$$f(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

Differentiating w.r.t. x , we get

$$f'(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \left[-\left(\frac{x-\mu}{\sigma} \right) \right] = -\frac{x-\mu}{\sigma} \cdot f(x)$$

$$\text{Now } f'(x) = 0 \Rightarrow x - \mu = 0 \quad \text{i.e., } x = \mu$$

$$f''(x) = -\frac{1}{\sigma^2} [(x-\mu) \cdot f'(x) + f(x)]$$

$$= -\frac{1}{\sigma^2} \left[(x-\mu) \cdot -\frac{(x-\mu)}{\sigma^2} \cdot f(x) + f(x) \right]$$

$$= \frac{-f(x)}{\sigma^2} \left[1 - \frac{(x-\mu)^2}{\sigma^2} \right]$$

At the point $x = \mu$, we have

$$f''(x) = -\left[\frac{f(x)}{\sigma^2} \right]_{x=\mu} = -\frac{1}{\sigma^2 \cdot \sqrt{2\pi} \cdot \sigma} < 0$$

Hence $x = \mu$ is the mode of the Normal Distribution.

4. Median of Normal distribution

[JNTU 2004 S]

If M is the median of the normal distribution, we have

$$\int_{-\infty}^M f(x) dx = \frac{1}{2}$$

$$\text{i.e. } \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2}$$

$$\text{i.e. } \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx + \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2} \quad \dots (1)$$

$$\text{Consider } \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{Put } \frac{x-\mu}{\sigma} = z. \text{ Then } dx = \sigma dz$$

$$\begin{aligned} \therefore \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^0 e^{-\frac{z^2}{2}} \cdot \sigma dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{-\frac{z^2}{2}} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{z^2}{2}} dz \text{ (by symmetry)} = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{\frac{\pi}{2}} = \frac{1}{2} \quad \dots (2) \end{aligned}$$

$\therefore$ From (1) and (2), we have

$$\frac{1}{2} + \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 0 \Rightarrow \int_{\mu}^M e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 0$$

$$\Rightarrow \mu = M \left[\because \text{if } \int_a^b f(x) dx = 0 \text{ then } a = b, \text{ where } f(x) > 0 \right]$$

Hence for the Normal Distribution, Mean = Median

Note : From above, we notice that for the Normal distribution mean, median and mode coincide.

i.e., mean = median = mode

[JNTU 2004S, 2008S (Set No.1,2)]

Hence the distribution is symmetrical.

5. Mean deviation from the Mean for Normal Distribution

[JNTU 2001S, (A) Dec. 2009 (Set No. 4)]

By definition, Mean deviation (about mean)

$$\begin{aligned}
 &= \int_{-\infty}^{\infty} |x - \mu| f(x) dx \\
 &= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} |x - \mu| \cdot e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2} dx \\
 &= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |z| e^{-z^2/2} dz \quad \left[\text{Putting } \frac{x - \mu}{\sigma} = z \right] \\
 &= \frac{\sigma}{\sqrt{2\pi}} \cdot 2 \int_0^{\infty} |z| \cdot e^{-\frac{z^2}{2}} dz \quad [\because \text{Integrand is an even function}] \\
 &= \sqrt{\frac{2}{\pi}} \cdot \sigma \int_0^{\infty} z e^{-\frac{z^2}{2}} dz \quad [\because |z| = z \text{ in } [0, \infty)] \\
 &= \sqrt{\frac{2}{\pi}} \cdot \sigma \int_0^{\infty} e^{-t} dt \quad \left[\text{Putting } \frac{z^2}{2} = t \right] \\
 &= \sqrt{\frac{2}{\pi}} \cdot \sigma \left(-e^{-t} \right)_0^{\infty} = \sqrt{\frac{2}{\pi}} \cdot \sigma (0 + 1) \\
 &= \sigma \cdot \sqrt{\frac{2}{\pi}} = \frac{4\sigma}{5} \text{ (approximately)}
 \end{aligned}$$

Hence the mean deviation from the Mean for Normal distribution is equal to $\frac{4}{5}$ times of Standard deviation approximately.

3.13 CHIEF CHARACTERISTICS OF THE NORMAL DISTRIBUTION

[JNTU 2004]

1. The graph of the Normal distribution $y = f(x)$ in the xy -plane is known as the normal curve.

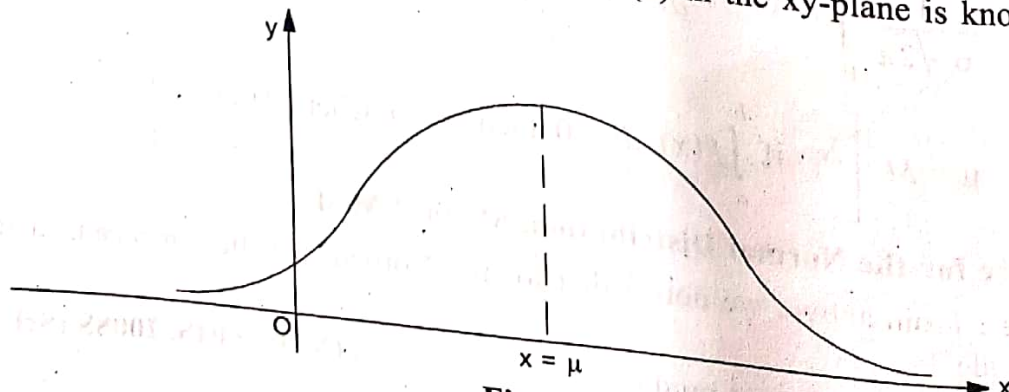
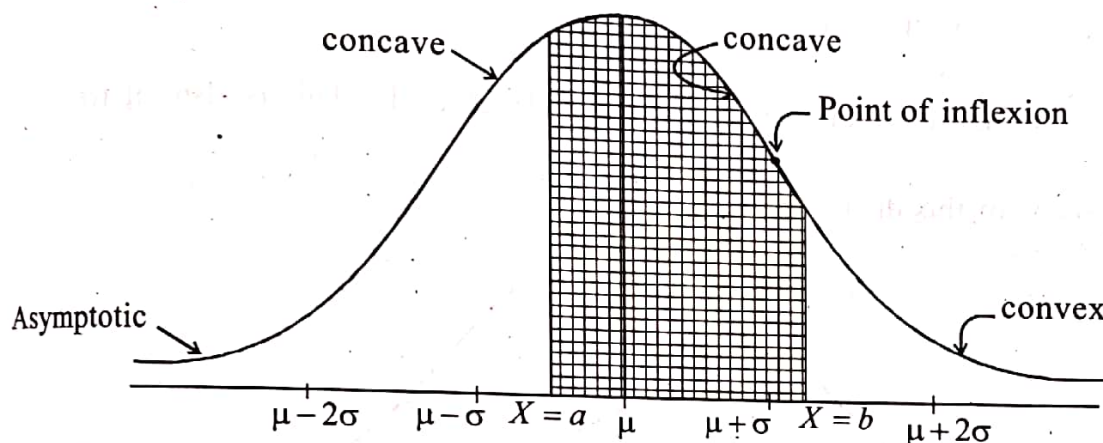


Fig.

2. The curve is a bell shaped curve and symmetrical with respect to mean *i.e.*, about the line $x = \mu$ and the two tails on the right and the left sides of the mean (μ) extends to infinity. The top of the bell is directly above the mean μ .
3. Area under the normal curve represents the total population.
4. Mean, median and mode of the distribution coincide at $x = \mu$ as the distribution is symmetrical. So normal curve is unimodal (has only one maximum point).
5. x -axis is an asymptote to the curve.
6. Linear combination of independent normal variates is also a normal variate.
7. The points of inflexion of the curve are at $x = \mu \pm \sigma$ and the curve changes from concave to convex at $x = \mu + \sigma$ to $x = \mu - \sigma$.



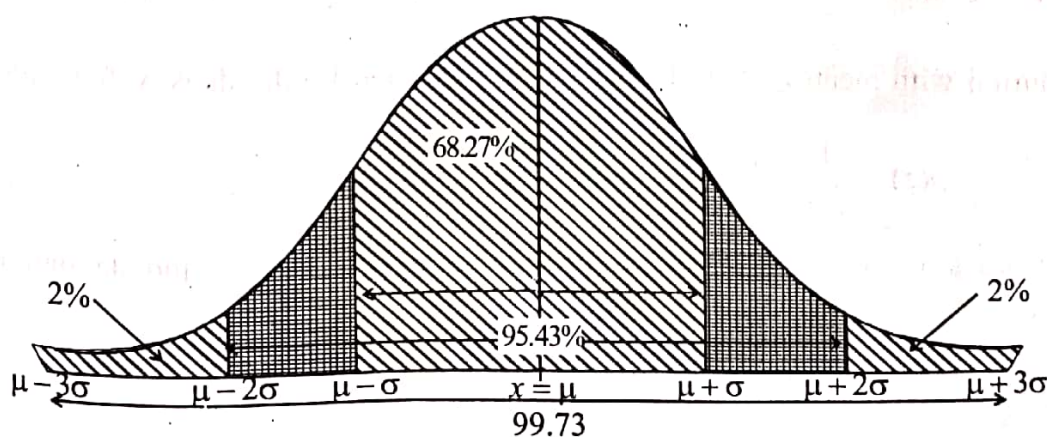
8. The probability that the normal variate X with mean μ and standard deviation σ lies between x_1 and x_2 is given by

$$P(x_1 \leq X \leq x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \quad \dots (1)$$

Since (1) depends on the two parameters μ and σ , we get different normal curves for different values of μ and σ and it is an impracticable task to plot all such

normal curves. Instead, by putting $z = \frac{x-\mu}{\sigma}$, the R.H.S. of equation (1) becomes independent of the two parameters μ and σ . Here z is known as the standard variable.

9. Area under the normal curve is distributed as follows :



- (i) Area of normal curve between $\mu - \sigma$ and $\mu + \sigma$ is 68.27%
i.e., $P(\mu - \sigma < X < \mu + \sigma) = 0.6826$.
- (ii) Area of normal curve between $\mu - 2\sigma$ and $\mu + 2\sigma$ is 95.43%.
- (iii) Area of normal curve between $\mu - 3\sigma$ and $\mu + 3\sigma$ is 99.73%.

Standard Normal Distribution :

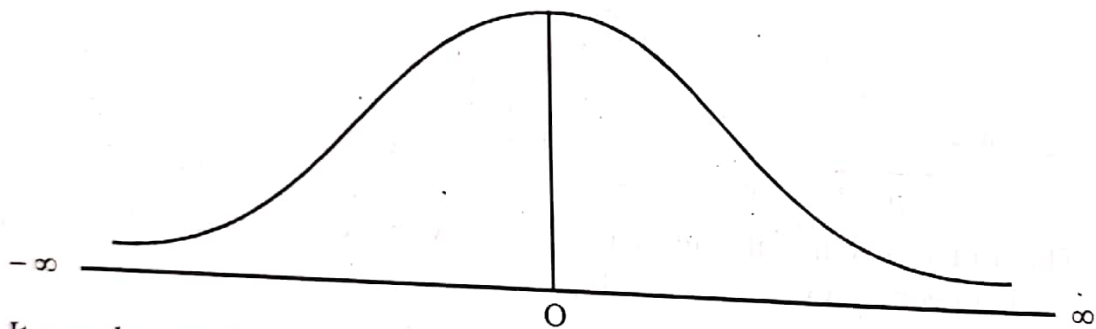
Definition : The Normal distribution with mean $(\mu) = 0$ and S. D. $(\sigma) = 1$, is known as *Standard Normal Distribution*.

The random variable that follows this distribution is denoted by z . If a variable x follows normal distribution with mean μ and s.d. σ , the variable z defined as

$$z = \frac{x - \mu}{\sigma}$$

has standard normal distribution with mean 0 and s.d. as 1. This is also referred as z -score.

The shape of this distribution is as follows:



It may be noted that the curve has the same shape as the normal distribution but, with the special properties that $\mu = 0$ and $\sigma = 1$ and has maximum at $z = 0$, and extends from $-\infty$ to $+\infty$ like normal distribution.

This is not the case with Binomial or Poisson distributions. They have different skewness (asymmetry) for different parameters.

We can transform any normal distribution to the standardised form because the normal distribution has the same shape whatever its parameter (μ and σ) may be.

If we take $\frac{x - \mu}{\sigma} = z$ in $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, we obtain the Standard Normal Distribution with mean zero and S. D. $(\sigma) = 1$ defined by the density function

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-(z^2/2)}, \quad -\infty < z < \infty$$

Here z is called the standard normal variate with mean 0 and standard deviation 1 and in short, we say that $p(z)$ is $N(0,1)$.

Area under any normal curve is found from the table of standard normal probability distribution showing the area between the mean and any value of the normally distributed random variable. For given values of μ and σ , and a specific value, x , of the random variable, the standardized variate z is derived from the formula $z = \frac{x - \mu}{\sigma}$. The purpose of standardisation of the normal distribution is that we can make use of the tables of the area of the standard curve $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ (representing probability) for the various points along the x -axis.

(The standard normal tables are given at the end of the book).

The standard normal distribution is also known as Unit Normal Distribution or Z-Distribution.

The standard normal curve helps us to find the areas within two assigned limits under the curve. The areas between the standard normal curve drawn at two assigned limits a and b will give the proportion of cases for which the values of z lie between a and b . Thus the area between two assigned limits a and b under the standard normal curve will represent the probability that z will lie between a and b . It is denoted by $P(a \leq z \leq b)$.

Uses of Normal Distribution:

1. The normal distribution can be used to approximate Binomial and Poisson distributions.
2. It has extensive use in sampling theory. It helps us to estimate parameter from statistic and to find confidence limits of the parameter.
3. It has a wide use in testing Statistical Hypothesis and Tests of significance in which it is always assumed that the population from which the samples have been drawn should have normal distribution.
4. It serves as a guiding instrument in the analysis and interpretation of statistical data.

Some other uses and situations where normal distribution is applicable are mentioned in the section 3.15.

3.14 AREA UNDER THE NORMAL CURVE (NORMAL PROBABILITY INTEGRAL) CONVERSION INTO STANDARD NORMAL FORM

By taking $z = \frac{x - \mu}{\sigma}$, the standard normal curve is formed. The total area under this curve is 1. The area under the curve is divided into two equal parts by $z = 0$. Left hand side area and right hand side area to $z = 0$ is 0.5. The area between the ordinate $z = 0$ and any other ordinate can be noted from the table given in the subsequent pages.

The probability that random variable X will lie between $X = \mu$ and $X = x_1$ is given by

$$P(\mu < X < x_1) = \int_{\mu}^{x_1} f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^{x_1} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_0^{z_1} e^{-\frac{z^2}{2}} \cdot \sigma dz$$

$$[\text{Putting } \frac{x - \mu}{\sigma} = z \text{ and } \frac{x_1 - \mu}{\sigma} = z_1]$$

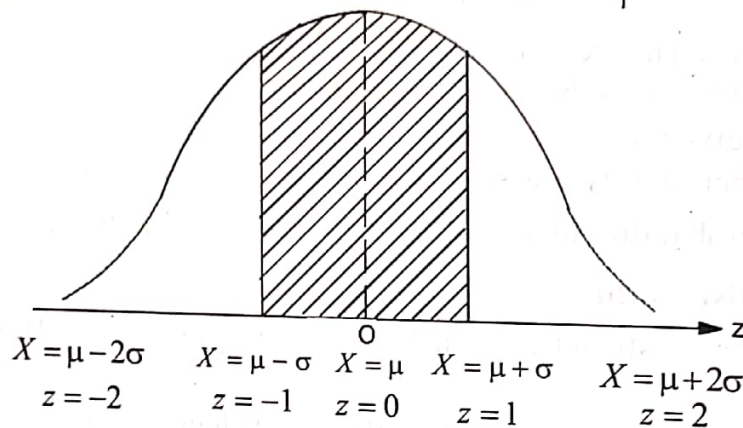
$$\text{i.e., } P(\mu < X < x_1) = P(0 < z < z_1) = \frac{1}{\sqrt{2\pi}} \int_0^{z_1} e^{-\frac{z^2}{2}} dz = A(z_1) \text{ say.}$$

$$\text{i.e., } A(z_1) = \frac{1}{\sqrt{2\pi}} \int_0^{z_1} e^{-\frac{z^2}{2}} dz.$$

The function $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$ is the probability function of standard normal variate.

The definite integral $\int_0^{z_1} e^{-\frac{z^2}{2}} dz$ is known as the *normal probability integral* and gives

the area under standard normal curve between $z = 0$ and $z = z_1$.



In particular, the probability that a random variable X lies in the interval $(\mu - \sigma, \mu + \sigma)$ is given by

$$P(\mu - \sigma < X < \mu + \sigma) = \int_{\mu - \sigma}^{\mu + \sigma} f(x) dx$$

$$\begin{aligned} \Rightarrow P(-1 < z < 1) &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{-\frac{z^2}{2}} dz \left(z = \frac{x - \mu}{\sigma} \right) \\ &= \frac{2}{\sqrt{2\pi}} \int_0^1 e^{-\frac{z^2}{2}} dz = 2 \int_0^1 \phi(z) dz \text{ where } \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \\ &= 2(0.3413) \text{ (from tables)} \\ &= 0.6826 \end{aligned}$$

Note : Since the integrand is an even function, it follows that $A(-z_1) = A(z_1)$.

$$\begin{aligned} \text{Similarly } P(\mu - 2\sigma < X < \mu + 2\sigma) &= P(-2 < z < 2) = \int_{-2}^2 \phi(z) dz \\ &= 2 \int_0^2 \phi(z) dz = 2(0.4772) = 0.9544 \end{aligned}$$

How to find Probability Density of normal curve

The probability that the normal variate X with mean μ and standard deviation σ lies between two specific values x_1 and x_2 with $x_1 \leq x_2$ can be obtained using area under the standard normal curve as follows :

Step 1 : Perform the change of scale $z = \frac{x - \mu}{\sigma}$ and find z_1 and z_2 corresponding to the values of x_1 and x_2 respectively.

Step 2 (a) : To find $P(x_1 \leq X \leq x_2) = P(z_1 \leq z \leq z_2)$

Case 1 : If both z_1 and z_2 are positive (or both negative), then

$$\begin{aligned} P(x_1 \leq X \leq x_2) &= |A(z_2) - A(z_1)| \\ &= (\text{Area under the normal curve from } 0 \text{ to } z_2) \\ &\quad - (\text{Area under the normal curve from } 0 \text{ to } z_1) \end{aligned}$$

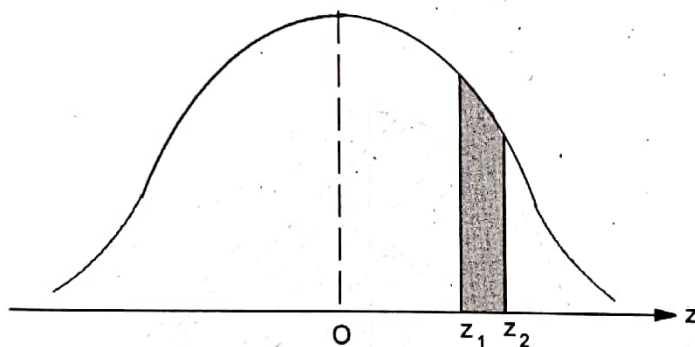


Fig.

Case 2 : If both $z_1 < 0$ and $z_2 > 0$ are positive, then

$$P(x_1 \leq X \leq x_2) = A(z_2) + A(z_1)$$

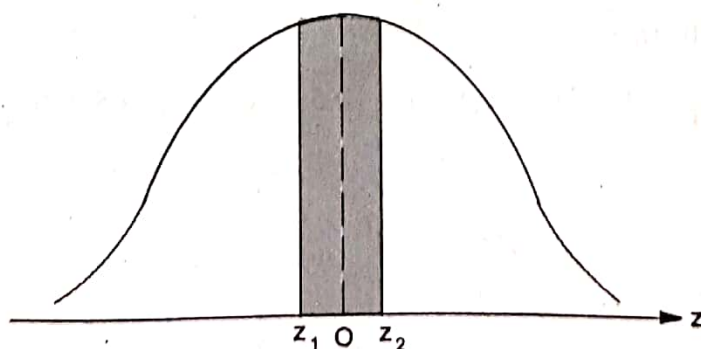


Fig.

Step 2 (b) : To find $P(z > z_1)$

Case 1 : If $z_1 > 0$ then

$$P(z > z_1) = 0.5 - A(z_1) \quad [\because P(z < 0) = P(z > 0) = \frac{1}{2}]$$

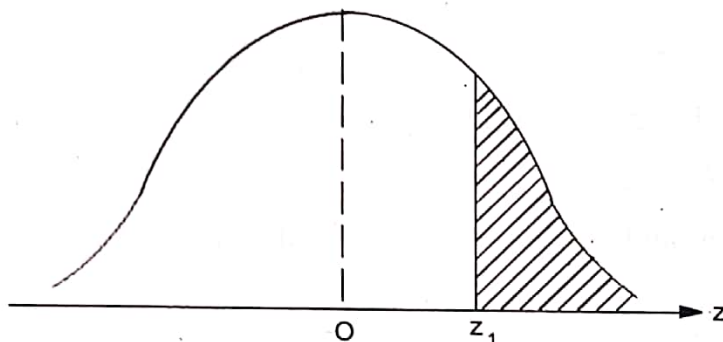


Fig.

Case 2 : If $z_1 < 0$, then $P(z > z_1) = 0.5 + A(z_1)$

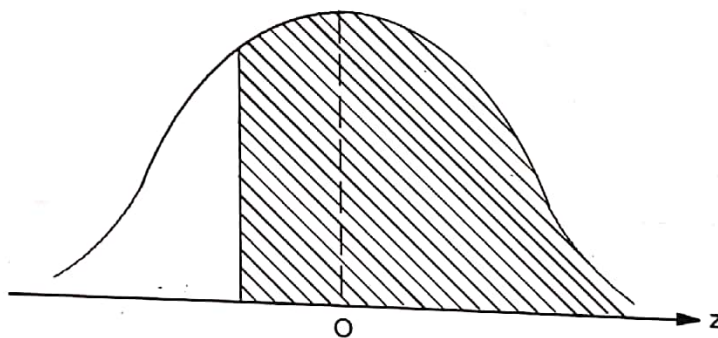


Fig.

Step 2 (c) : To find $P(z < z_1) = 1 - P(z > z_1)$

Case 1 : If $z_1 > 0$, then

$$P(z < z_1) = 1 - P(z > z_1) = 1 - [0.5 - A(z_1)] = 0.5 + A(z_1)$$

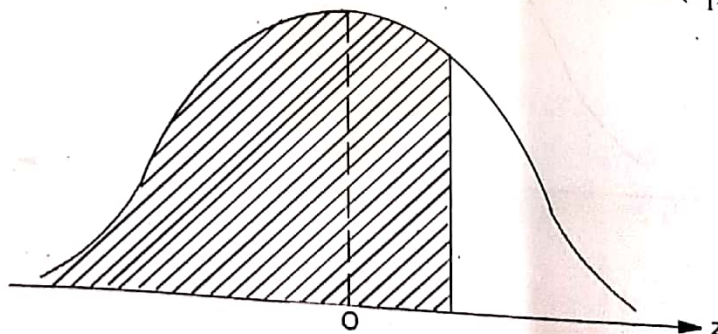


Fig.

Case 2 : If $z_1 \leq 0$, then

$$P(z < z_1) = 1 - P(z > z_1) = 1 - [0.5 + A(z_1)] \\ = 0.5 - A(z_1)$$

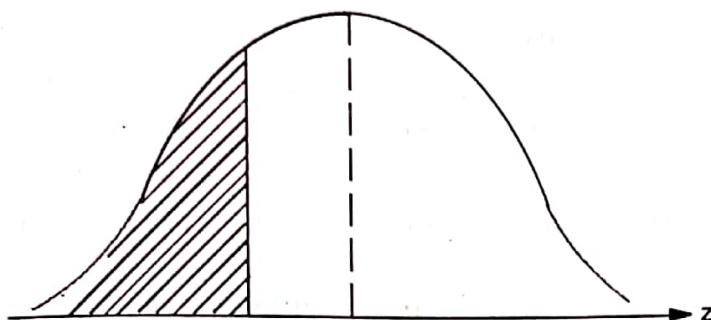


Fig.

3.15 THE IMPORTANCE AND APPLICATIONS OF THE NORMAL DISTRIBUTION

[JNTU 2000, 200S, 20005 (Set No. 4)]

Normal distribution plays a very important role in statistical theory because of the following reasons :

1. Data obtained from Psychological, Physical and Biological measurements approximately follow Normal distribution. I.Q. scores, heights and weights of individuals etc., are examples of measurements which are normally distributed or nearly so.
2. Most of the distributions that are encountered in practice, for example, Binomial, Poisson, Hypergeometric, etc. can be approximated to Normal distribution. If the number of trials n is indefinitely large and neither p nor q is very small, then Binomial distribution tends to Normal distribution. If the parameter $\lambda \rightarrow \infty$, then Poisson distribution tends to Normal distribution.
3. Since the Normal distribution is a limiting case of the Binomial distribution for exceptionally large numbers, it is applicable to many applied problems in kinetic theory of gases and fluctuations in the magnitude of an electric current.
4. Even if a variable is normally distributed, it can sometimes be brought to normal form by simple transformation of the variable.
5. For large samples, any statistic (i.e., sample mean, sample S.D., etc.) approximately follows Normal distribution and as such it can be studied with the help of normal curve.
6. Normal curve is used to find confidence limits of the population parameters.
7. The proofs of all the tests of significance in sampling are based upon the fundamental assumption that the population from which the samples have been drawn is normal.
8. Normal distribution finds large applications in Statistical Quality Control in industry for finding control limits.